

Chapter 3 Defects in Ceramics

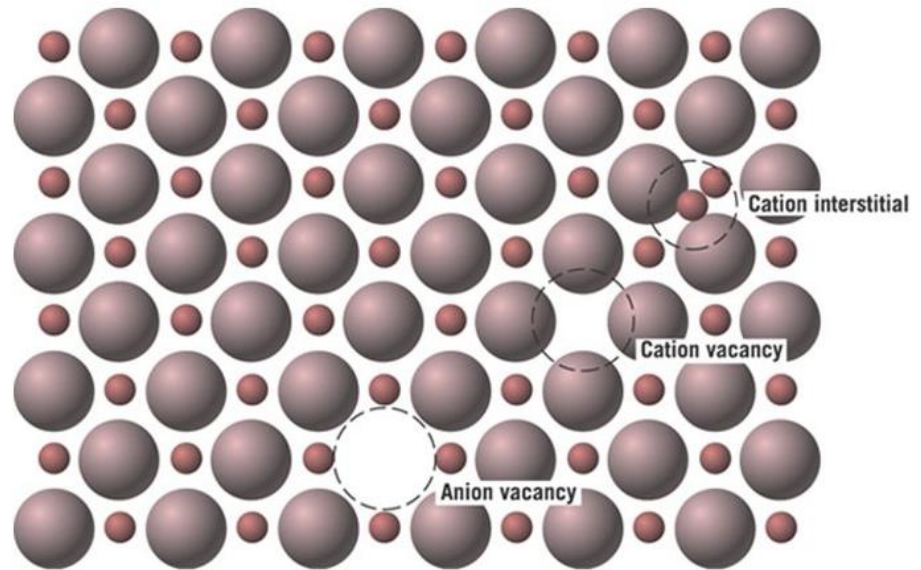
A great many properties of crystals are determined by **imperfections**.

- Electrical conductivity
- Diffusion transport
- Optical properties
- Rate of kinetic process

Kinetic processes

- Precipitation
- Densification
- Grain coarsening
- High temperature creep deformation

imperfection



Point Defects

✦ Point defects : defined as deviations from the perfect atomic arrangement.

- Missing ions
- Substituted ions
- Interstitial ions
- Associated valence electrons

Point defects occur in all crystalline materials

Point Defects in Ceramics

Point defects

1) Ionic defects:

- vacancies
- interstitial
- substitutional

2) Electronic defects: formed when valence electrons are excited into higher orbital energy levels.

Such an excitation may create an electron in the conduction band and/or an electron hole in the valence band of the crystal.

How Point Defects in Ceramics are Formed?

Point defects in ceramics can be formed by

- thermal excitation
- solutes (impurities)
- oxidation or reduction

➤ In the solid-state analogy, the perfect crystal may be regarded as a neutral medium into which the charged defects are dissolved.

➤ Solid-state defect interaction $\longleftrightarrow$ similar $\longleftrightarrow$ Solution-chemical interaction.
(defect chemistry)

Point defect behavior

Two principle issues:

- What types of point defects are present ?
- In what concentration ?

Depend on:

- 1)Crystal structure;
- 2)Chemical composition;
- 3)Bonding;
- 4)Temperature

Physical Properties provides strong clues to defect behavior

- 1)density; 2) melting point; 3)electrical conductivity;
- 4) diffusion; 5) optical absorption

The importance of melting temperature

- The formation of atomic defects requires the breaking of interatomic bonds
- The melting temperature (T_m) of a compound scale with the strength of its interatomic bonding.

• T/T_m
(homologous temperature) $\longleftrightarrow$ *scale to* $\left\{ \begin{array}{l} \text{defect} \\ \text{diffusion} \end{array} \right\}$ properties

The importance of T_m and crystal structure

- Ex :

At equal temperature, the conc. of defects in MgO is many orders of magnitude lower than NaCl.
MgO ($T_m = 2825\text{ }^\circ\text{C}$), NaCl ($T_m < 1000\text{ }^\circ\text{C}$)
- At equal homologous temperature, the defect conc. are rather similar.

Defect sites in the crystal structure
- The availability of defect sites in the crystal structure

A close – packed sublattice of ions does not easily accommodate interstitial defect

The availability of defect sites in the crystal structure

Ex : rocksalt structure

Anion interstitials > $\begin{cases} \text{anion vacancies} \\ \text{cation vacancies} \\ \text{cation interstitials} \end{cases}$ (high energy)

- The location of solutes and impurities in the lattice often depends on the compatibility in $\begin{cases} \text{size} \\ \text{valence} \end{cases}$ with host ions.
- Solutes that differ in valence from the host (aliovalent) must be compensated by additional charged defects in order to maintain overall electrical neutrality in the crystal

Nonstoichiometric composition

- Nonstoichiometric compounds are those in which the metal/anion ratio deviates from the ideal value on which the structure is based, due to the existence of multiple ion valence state.
- Transition metal oxides (NiO , FeO , TiO_2 ...) are often highly defective.

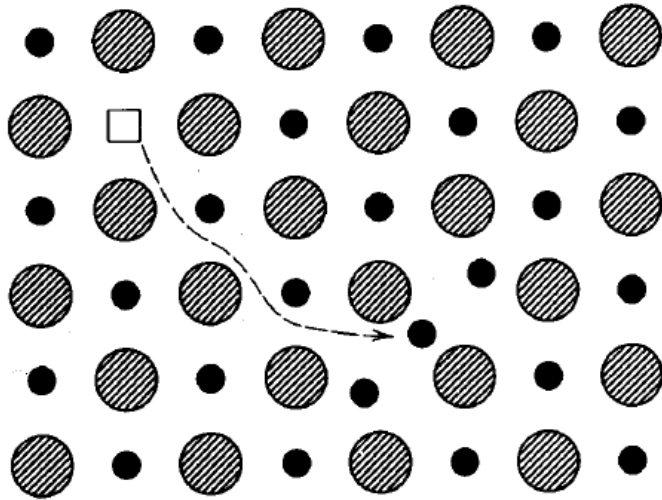


- Cation-deficient due to the presence of a significant fraction
(at least 5%) of the iron being in the Fe^{3+} state

Intrinsic ionic disorder

- Crystalline defects in ionic materials $\left\{ \begin{array}{l} \text{Frenkel} \\ \text{Schottky} \end{array} \right.$ defects
- Intrinsic defects, they can be thermally generated in perfect crystal.
- Extrinsic defects, which are formed only by the addition of impurities or solute.

Frenkel disorder



Frenkel disorder. Ion leaving normal site forms an interstitial ion and leaves a vacancy.

- A Frenkel defect is formed when an atom is displaced from its normal site onto an interstitial site forming a

defect pair { a vacancy
an interstitial }

- In ionic materials, cation / anion can undergo this kind of displacement.
- In { metals
covalent compounds , } Frenkel defects can also form.
- They differ from those in ionic compounds only in that the defect need not be electrically charged .

Schottky disorder

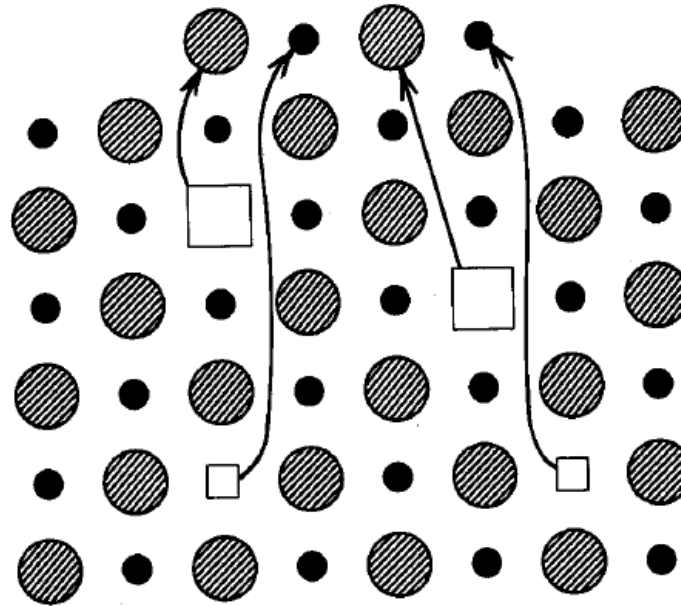


Fig. 2.2 Schottky disorder. Displacement of anion and cation to surface leaves a pair of vacancies.

- The Schottky defect is unique to ionic compound and is represented by the simultaneous creation of both $\left\{ \begin{array}{l} \text{cation} \\ \text{anion} \end{array} \right\}$ vacancies.
- The vacancies must be formed in the stoichiometric ratio in order to preserve the electrical neutrality of the crystal.

Kröger–Vink notation

Kröger–Vink notation is a set of conventions that are used to describe electric charge and lattice position for point defect species in crystals.

The notation follows the scheme: M_S^C

M corresponds to the species. These can be

- atoms – e.g., Si, Ni, O, Cl,
- vacancies – V or v (since V is also the symbol for vanadium)
- interstitials – i
- electrons – e
- electron holes – h

S indicates the lattice site that the species occupies.

C corresponds to the electronic charge of the species relative to the site that it occupies. The charge of the species is calculated by the charge on the current site minus the charge on the original site.

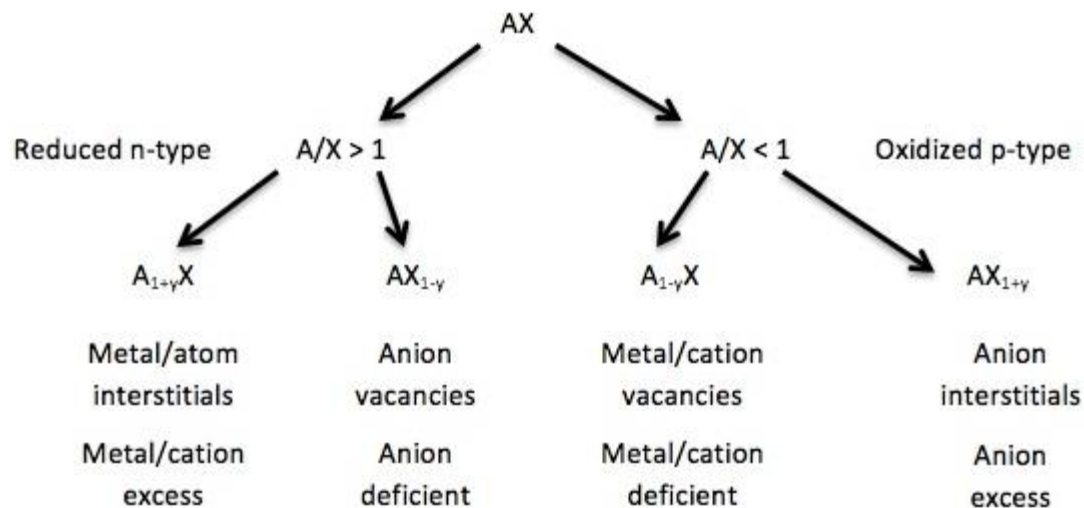
× is used to indicate null charge.

- indicates a net single positive charge, while two would represent two net positive charges;
- ' signifies a net single negative charge, so two would indicate a net double negative charge.

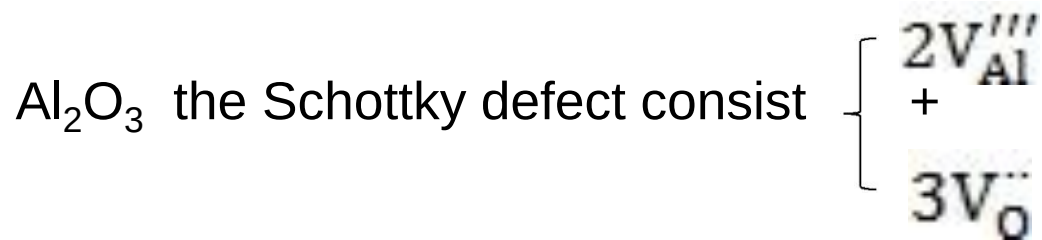
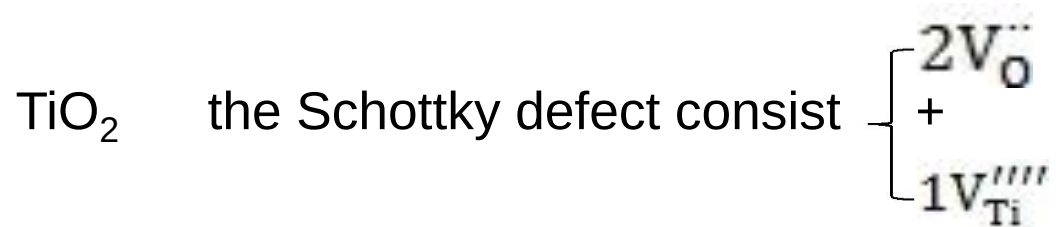
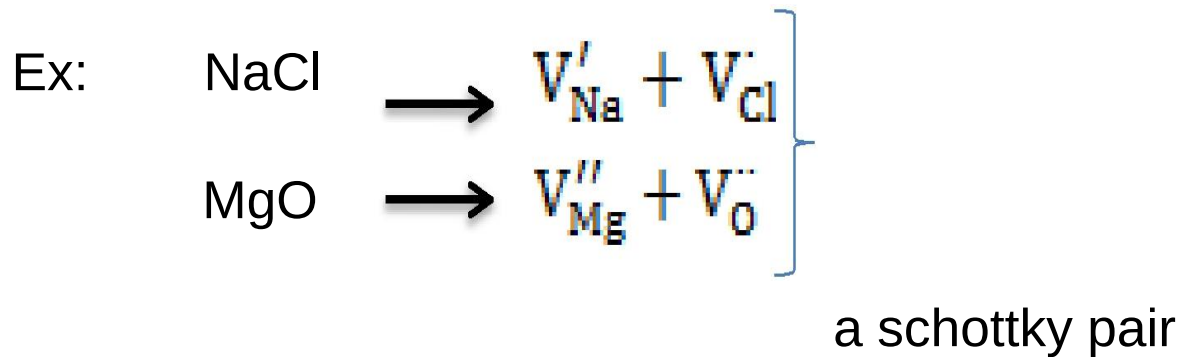
Kröger–Vink notation

- $\text{Al}_{\text{Al}}^{\times}$ — an aluminum ion sitting on an aluminum lattice site, with neutral charge.
- $\text{Ni}_{\text{Cu}}^{\times}$ — a nickel ion sitting on a copper lattice site, with neutral charge.
- $\text{v}_{\text{Cl}}^{\bullet}$ — a chlorine vacancy, with single positive charge.
- $\text{Ca}_i^{\bullet\bullet}$ — a calcium interstitial ion, with double positive charge.
- Cl_i' — a chlorine anion on an interstitial site, with single negative charge.
- O_i'' — an oxygen anion on an interstitial site, with double negative charge.
- e^- — an electron. No site is normally specified.

Oxidation–reduction tree

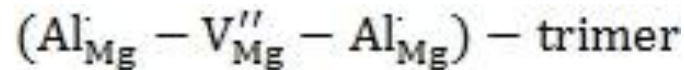
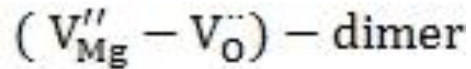


Kröger–Vink notation Examples



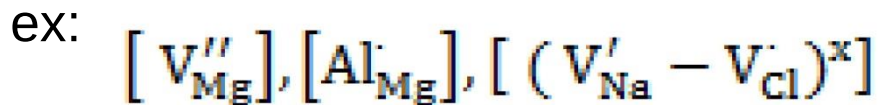
Kröger–Vink notation

- Cluster defects or defect associates



The concentration of defects is denoted by [____]

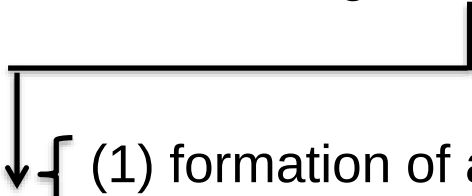
└→ square brackets



- Isovalent substitution, NiO in MgO



Aliovalent substitution, -greater or lesser in valence than the host, must be charge compensated in solid solution.

- 
- ↓ { (1) formation of additional ionic defect $\longrightarrow$ ionic compensation
 - (2) liberating electrons and holes $\longrightarrow$ electronic compensation

Concentration of intrinsic defects

Consider the change in free energy of a perfect crystal with initial free energy G_0 , upon forming n Frenkel defect pairs at an energy of Δg_f per pair. The free energy of the crystal becomes:

$$G = G_0 + n\Delta g_f - T\Delta S_c$$

(ΔS_c : configurational entropy of the crystal)

The change in free energy

$$\Delta G = (G - G_0) = n\Delta g_f - T\Delta S_c$$

$$\Delta S_c = k \ln \Omega$$

(Ω is the number of distinct ways in which the defect can be arranged)

Concentration of intrinsic defects

(1) If these are arranged on a total of N lattice site, the vacancies can be arranged in Ω_v ways.

$$\Omega_v = \frac{N!}{(N - n_v)! n_v!}$$

(n_v : number of vacancies
 n_i : number of interstitial)

(2) If these are arranged on a total of N lattice site, the interstitial can be arranged in Ω_i ways.

$$\Omega_i = \frac{N!}{(N - n_i)! n_i!}$$

The total number of configuration:

$$\Omega = \Omega_i \Omega_v$$

$$\because n_v = n_i = n$$

$$\Delta S_c = k \ln \left[\frac{N!}{(N - n_v)! n_v!} \right] \left[\frac{N!}{(N - n_i)! n_i!} \right]$$

$$= 2k \ln \left[\frac{N!}{(N - n)! n!} \right]$$

$$\approx 2k [N \ln N - (N - n) \ln (N - n) - n \ln n]$$

(stirling' s approximation $\ln N! = N \ln N - N$)

The total free energy change:

$$\Delta G = n \Delta g - 2kT \left[N \ln \left(\frac{N}{N - n} \right) + n \ln \left(\frac{N - n}{n} \right) \right]$$

- To find the equilibrium conc. of defects.

$$\left(\frac{\partial \Delta G}{\partial n}\right)_{T,P} = 0 \quad (\text{for dilute conc. of defect } (N - n) \approx N)$$

For the conc. of defects:

$$\frac{n}{N} = \exp\left(-\frac{\Delta g}{2kT}\right) = \exp\left(\frac{\Delta s}{2k}\right) \exp\left(-\frac{\Delta h}{2kT}\right)$$

ΔS : non – configuration entropy

Δh : energy of formation

$\Delta S \ll$ Configuration entropy

$\exp\left(\frac{\Delta s}{2kT}\right)$ rangin from $10^{-4} \sim 10^4$ (from experimental data)

Defect concentration at different temperatures

- In table 2.1, defect conc. are shown as a function of h and temperature, assuming that $\exp(\Delta S/2k) \sim 1$.

Table 2.1 Defect Concentration at Different Temperatures

$$\frac{n}{N} = \exp\left[-\frac{\Delta g}{2kT}\right] = \exp\left[\frac{\Delta s}{2k}\right] \exp\left[-\frac{\Delta h}{2kT}\right] \approx \exp\left(-\frac{\Delta h}{2kT}\right)$$

Defect Concentration	1eV ^a	2eV	4eV	6eV	8eV
n / N at 100°C	2×10^{-7}	3×10^{-14}	1×10^{-27}	3×10^{-41}	1×10^{-54}
n / N at 500°C	6×10^{-4}	3×10^{-7}	1×10^{-13}	3×10^{-20}	8×10^{-27}
n / N at 800°C	4×10^{-3}	2×10^{-5}	4×10^{-10}	8×10^{-15}	2×10^{-19}
n / N at 1000°C	1×10^{-2}	1×10^{-4}	1×10^{-8}	1×10^{-12}	1×10^{-16}
n / N at 1200°C	2×10^{-2}	4×10^{-4}	1×10^{-7}	5×10^{-11}	2×10^{-19}
n / N at 1500°C	4×10^{-2}	1×10^{-4}	2×10^{-6}	3×10^{-9}	4×10^{-12}
n / N at 1800°C	6×10^{-2}	4×10^{-3}	1×10^{-5}	5×10^{-8}	2×10^{-10}
n / N at 2000°C	8×10^{-2}	6×10^{-3}	4×10^{-5}	2×10^{-7}	1×10^{-9}

^a1eV = 23.05 kcal/mole.

Intrinsic V.S. Extrinsic Behavior

- The concentration of intrinsic defect can be small in highly refractory ceramics of large defect formation energy.

Intrinsic defect – T

Extrinsic defect – $\left\{ \begin{array}{l} \text{solute (especially aliovalent solutes)} \\ \text{oxidation process} \\ \text{reduction process} \end{array} \right\} \text{process}$

- Intrinsic defect concentrations increase with temperature.
- Extrinsic defects with the exception of nonstoichiometry, concentration is constant.

Ex. Rocksalt structure type – NaCl and MgO

- In both, the dominant intrinsic defect is the Schottky defect.
- The concentration of defects

$$\frac{n}{N} = \exp\left(\frac{\Delta S}{2k}\right) \exp\left(-\frac{\Delta h}{2kT}\right)$$
$$= \exp\left(-\frac{\Delta g}{2kT}\right)$$

Δh : Energy of formation
 Δg : Formation free energy
 k : Boltzman constant

$$\Delta h (\text{NaCl}) = 2.2 \sim 2.4 \text{ eV}, \quad T_m = 801^\circ\text{C}$$

$$\Delta h (\text{MgO}) = 7.74 \text{ eV}, \quad T_m = 2825^\circ\text{C}$$

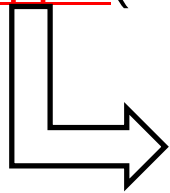
- High Δh , with the stronger bonding and higher T_m .
- At 700°C , n_v/N (MgO) has many orders of magnitude fewer than n_v/N (NaCl).

But at $\frac{T}{T_m} = 1$,

$$\frac{n_v}{N}(\text{NaCl}) \cong 2 \times 10^{-6} \text{ (2ppm)}$$
$$\frac{n_v}{N}(\text{MgO}) \cong 4 \times 10^{-7} \text{ (0.4ppm)}$$

- There is a great difference in the purity level with zone-refining procedures,

NaCl ~ 1ppm(impurities) } aliovalent cations.
MgO ~ 50ppm(impurities) }



{ (Because of the much higher processing temperatures necessary, contamination from crucibles is harder to avoid.)

- MgO, the impurity concentration, is much greater than the intrinsic defect concentration.
- NaCl, intrinsic defect can in practice be achieved rather easily.

Defect formation energy

Table 2.2 Some Defect Energies of Formation

Compound	Reaction	Energy of Formation Δh (eV)
AgCl	$\text{Ag}_{\text{Ag}}^x \rightleftharpoons \text{Ag}_i + V_{\text{Ag}}'$	1.1
NaCl	$\text{null} \rightleftharpoons V_{\text{Na}}' + V_{\text{Cl}}'$	2.2–2.4
KCl	$\text{null} \rightleftharpoons V_{\text{K}}' + V_{\text{Cl}}'$	2.6
LiF	$\text{null} \rightleftharpoons V_{\text{Li}}' + V_{\text{F}}'$	2.4–2.7
CsCl	$\text{null} \rightleftharpoons V_{\text{Cs}}' + V_{\text{Cl}}'$	1.86
BeO	$\text{null} \rightleftharpoons V_{\text{Be}}'' + V_{\text{O}}'$	~ 6
MgO	$\text{null} \rightleftharpoons V_{\text{Mg}}'' + V_{\text{O}}''$	7.7
CaO	$\text{null} \rightleftharpoons V_{\text{Ca}}'' + V_{\text{O}}''$	~ 6
BaO	$\text{null} \rightleftharpoons V_{\text{Ba}}'' + V_{\text{O}}''$	3.4
MnO	$\text{null} \rightleftharpoons V_{\text{Mn}}'' + V_{\text{O}}''$	4.6
FeO	$\text{null} \rightleftharpoons V_{\text{Fe}}'' + V_{\text{O}}''$	6.5
ZnO	$\text{O}_{\text{O}}^x \rightleftharpoons \text{O}_i'' + V_{\text{O}}''$	2.51
Li ₂ O	$\text{Li}_{\text{Li}}^x \rightleftharpoons \text{Li}_i + V_{\text{Li}}'$	2.28
CaF ₂	$\text{F}_{\text{F}}^x \rightleftharpoons V_{\text{F}}' + \text{F}_i'$	2.3–2.8
	$\text{Ca}_{\text{Ca}}^x \rightleftharpoons V_{\text{Ca}}'' + \text{Ca}_i''$	~ 7
	$\text{null} \rightleftharpoons V_{\text{Ca}}'' + 2 V_{\text{F}}'$	~ 5.5
UO ₂	$\text{O}_{\text{O}}^x \rightleftharpoons \text{O}_i'' + V_{\text{O}}''$	5.1
	$\text{U}_{\text{U}}^x \rightleftharpoons V_{\text{U}}''' + \text{U}_i''''$	~ 9.5
	$\text{null} \rightleftharpoons V_{\text{U}}''' + 2 V_{\text{O}}''$	~ 6.4
TiO ₂ (rutile)	$\text{null} \rightleftharpoons V_{\text{Ti}}''' + 2 V_{\text{O}}''$	5.2
	$\text{O}_{\text{O}}^x \rightleftharpoons \text{O}_i'' + V_{\text{O}}''$	8.7
	$\text{Ti}_{\text{Ti}}^x \rightleftharpoons \text{Ti}_i'''' + V_{\text{Ti}}'''$	12
α -Al ₂ O ₃	$\text{null} \rightleftharpoons 2 V_{\text{Al}}''' + 3 V_{\text{O}}''$	20.1–25.7 (4.2–5.1 eV/defect)
	$\text{Al}_{\text{Al}}^x \rightleftharpoons \text{Al}_i'''' + V_{\text{Al}}'''$	10.4–14.2 (5.2–7.1 eV/defect)
	$\text{O}_{\text{O}}^x \rightleftharpoons \text{O}_i'' + V_{\text{O}}''$	7.6–14.5 (3.8–8.3 eV/defect)
MgAl ₂ O ₄	$\text{null} \rightleftharpoons V_{\text{Mg}}'' + 2 V_{\text{Al}}''' + 4 V_{\text{O}}''$	29.1 (4.15 eV/defect)

Units for defect concentration

- Units for defect concentration

$\frac{n_v}{N}$: mole fraction, # fraction of defect n , relative to # of possible site N .

n : number per unit volume (no./cm³ or cm⁻³)

N : density of atoms in solid ($\sim 10^{23}$ cm⁻³)

$$\text{factor : } \frac{Na(\text{no./mole})\rho(\text{g/cm}^3)}{\text{MW}(\text{g/mole})}$$

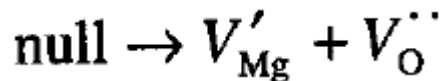
$$\text{mole fraction} \xrightarrow{\times f} \text{cm}^{-3}$$

$$Na \text{ (Avogadro' s no.)} \\ = 6.02 \times 10^{23}/\text{mole}$$

Defect chemical reactions

- From a statistical thermodynamic viewpoint, how defect concentration depend on $\left\{ \begin{array}{l} \text{formation energy } (\Delta h) \\ \text{temperature } (T) \end{array} \right\}$
- An equivalent way to view the formation of defects is as a chemical reaction, for which there is an equilibrium constant which is governed by the law of mass action.

Ex. Schottky reaction for NaCl and MgO



 (indicates the creation of defects from a perfect lattice.)

mass-action equilibrium constant

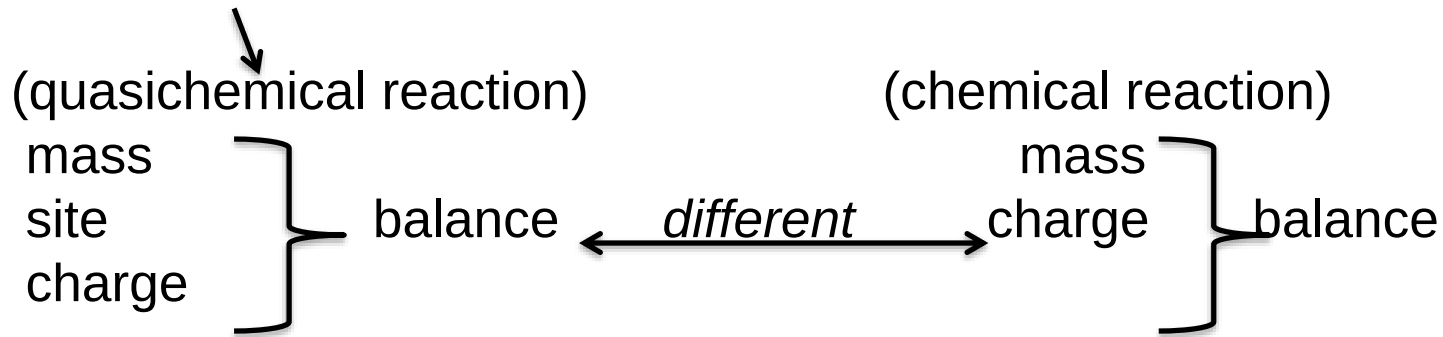
$$\left. \begin{aligned} k_S &= [V'_{Na}][V_{Cl}] \\ k_S &= [V''_{Mg}][V_{\dot{O}}] \end{aligned} \right\} \begin{aligned} &([] : \text{mole fraction } (\frac{n_v}{N}) \\ &\text{at fixed } T, \text{ vacancies conc.} \\ &\text{is a constant) } \end{aligned}$$

$$\begin{aligned} -k_S &= \exp\left[-\frac{\Delta g_S}{kT}\right] \\ k_S &= f(T) \end{aligned} \quad (\Delta g_S : \text{the formation free energy for Schottky pair.})$$

- When only the intrinsic defects are present,

$$[V'_{Na}] = [V_{Cl}] = \exp\left[-\frac{\Delta g_S}{2kT}\right]$$

Defect chemical reaction must obey



- mass balance : a chemical reaction cannot create or lose mass.
- site balance : the ratio of cation to anion site of the crystal must be preserved, although the total no. of sites can be increased or decreased.
- Charge balance : total effective charge is balanced.
(cation and anion vacancies must be formed in the stoichiometric ratios, the effective charges are automatically balanced.)

In addition to the formation of intrinsic defect, reaction of particular interest include :

- The incorporation of solutes
- Formation of intrinsic electron defects
- Oxidation/reduction
- Defect association and precipitation

Solute Incorporation

- Solute may enter solid solution in crystal as either substitutional or interstitial species

Ionic compensation mechanisms :

Ex1: The dissolution of Al_2O_3 in MgO

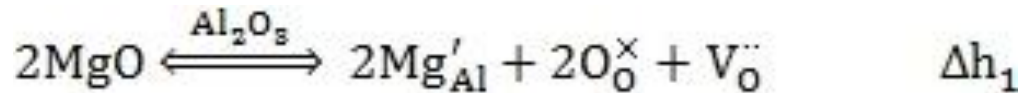
- Based on $r_{\text{Al}^{3+}} \approx r_{\text{Mg}^{2+}}$, we may presume that Al^{3+} will substitute for Mg^{2+} , the O ions are likely to occupy additional O lattice sites.



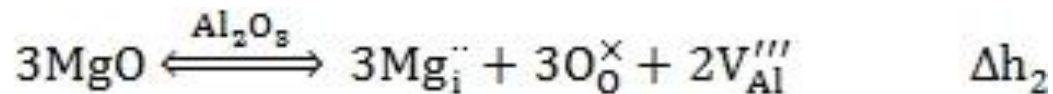
Ex2: The incorporation of MgO into Al_2O_3

- Mg ion may enter the solid solution { substitutionally.
interstitially.

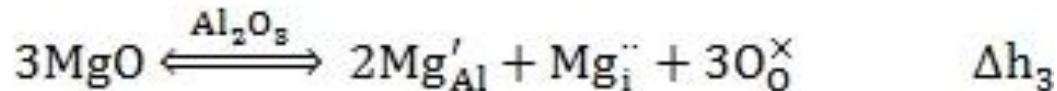
(1) If it is substitutional:



(2) If it is interstitial:



(3) Self-compensating (from both interstitial/substitutional defect)



If $\Delta h_1 \approx \Delta h_2 \approx \Delta h_3$, no single one will be the dominant mechanism of incorporation. (simultaneously in equilibrium)